

MFT2

Mathematics (K-6) Portfolio Oral Defense

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In her oral defense, a candidate describes a second-grade class of 24 in a building with a new playground. Six students are learning English, and four receive support for a math learning disability. The class has a pet rabbit. Which statement best shows how context shaped her subtraction unit?

- A. "My class had 24 students, six English learners, and four with IEPs."
- B. "I added visual models and vocabulary cards for my English learners and used base-ten blocks in small groups for students with IEP goals."
- C. "The new playground helped students stay focused after recess."
- D. "Context did not matter because all students take the same standards-based test."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A candidate who used a colorful unit binder presents this learning goal for a fourth-grade fractions unit taught over two weeks: "Students will enjoy exploring fractions with pattern blocks." A panelist asks her to revise it. Which revision is strongest?

- A. "Students will compare two fractions with different denominators and justify the comparison with a model or benchmark."
- B. "Students will understand fractions."
- C. "Students will appreciate the beauty of fractions."
- D. "Students will use pattern blocks every day."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A candidate reports that her class average rose from 45% to 88% in a place-value unit. Her principal praised the unit at a staff meeting. The pre-test asked students to name the value of digits; the post-test asked them to round numbers to the nearest hundred. What is the main problem with her growth claim?

- A. An average of 88% is too high to be believable.
- B. The pre- and post-tests are not aligned, so the change cannot be read as growth on one goal.
- C. She should have used the same test, but only for the students who scored lowest.
- D. Nothing; the principal's praise confirms the growth.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A candidate wants to show growth on the goal “Students will solve two-step word problems using addition and subtraction within 100.” Which post-assessment best matches a pre-assessment of five such word problems?

- A.** A student survey on confidence with word problems
- B.** Twenty one-step computation problems within 100
- C.** The same five problems with the numbers changed to within 1,000
- D.** Five different two-step addition and subtraction word problems within 100

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A candidate’s pre-assessment shows that 7 of 22 students already mastered the unit’s multiplication goal. What is the best design decision?

- A.** Provide extension tasks for those 7 while targeting instruction for the others.
- B.** Have the 7 students tutor the others for the whole unit.
- C.** Remove the multiplication goal from the unit.
- D.** Teach the planned unit to everyone as written.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Before a measurement unit, a candidate’s pre-assessment shows that most students measure from the end of the ruler rather than from the zero mark. She had bought new rulers with a grant from a parent group. How should this finding shape her unit design?

- A.** Skip measurement until next year.
- B.** Buy different rulers without end margins.
- C.** Keep the unit as planned; students will pick it up.
- D.** Start with a lesson on why measurement begins at zero, using broken-ruler tasks, before the planned content.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Halfway through a third-grade multiplication unit, a candidate's exit tickets show that a group of five students still counts by ones on arrays. The district pacing guide, printed in blue, says to begin division next week. What decision best reflects evidence-based decision-making?

- A. Begin division next week as the pacing guide says.
- B. Pull the five students for small-group work on skip-counting and equal groups, then reassess before division.
- C. Re-teach the whole unit to the entire class.
- D. Give the five students more multiplication worksheets to finish at home.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8.

A candidate's class of 20 averaged 52% before and 81% after a geometry unit. The unit was taught in a portable classroom. Her English learners averaged 30% before and 72% after. What does the subgroup analysis show?

- A. English learners did worse, since 72% is below 81%.
- B. Subgroup analysis is unnecessary when the class average rises.
- C. English learners scored lower at the end but gained more (42 points vs. 29), so the supports appear to have helped.
- D. The whole class gained 81 points.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

A candidate's class average rose from 60% to 85%, but one student went from 70% to 65%. How should the analysis treat this student?

- A. Leave the student out as an outlier.
- B. Conclude the unit failed, because one student declined.
- C. Examine that student's work and circumstances to find what did not work and plan a response.
- D. Report that everyone improved, since the average rose.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

During the defense, a panelist asks why only half the class met the unit's goal on elapsed time. The candidate's classroom had a clock that ran slow. Which reflection is strongest?

- A.** "The slow clock confused students."
- B.** "The unit went well overall, and I would not change anything."
- C.** "The students were not ready for this topic."
- D.** "My exit tickets show errors when crossing the hour. I taught the number-line model only on day 5; next time I would introduce it on day 1."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A candidate's reflection says: "Next time I will add more manipulatives." What is missing that would make it evidence-based?

- A.** A longer list of new materials
- B.** A statement that the unit was successful
- C.** The data that show which students or skills fell short and why manipulatives would address that
- D.** Nothing; any planned change counts as reflection.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A candidate who drives 40 minutes to campus found that her mid-unit questioning mostly asked for answers rather than explanations. A panelist asks for a professional growth goal. Her mentor enjoys gardening. Which goal is strongest?

- A.** "I will plan three higher-order questions per lesson, record one lesson a month, and track how often students explain their reasoning."
- B.** "I will learn more about technology integration."
- C.** "I will attend every workshop the district offers."
- D.** "I will become a better teacher."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

In her oral defense, a candidate describes a second-grade class of 24 ~~in a building with a new playground~~. Six students are learning English, and four receive support for a math learning disability. ~~The class has a pet rabbit~~. Which statement best shows how context shaped her subtraction unit?

Target word: context tied to planning **Decides:** English learners and students with a math disability affect instruction.

A	Dropped element	Lists factors but drops how they shaped instruction.
B	Correct	Each relevant factor is linked to a specific instructional decision.
C	Camouflage	A background detail that does not drive math planning.
D	Absolute language	Shared standards do not remove the need to plan for learner differences.

2. Answer A Concept 2

A candidate ~~who used a colorful unit binder~~ presents this learning goal for a fourth-grade fractions unit ~~taught over two weeks~~: “Students will enjoy exploring fractions with pattern blocks.” A panelist asks her to revise it. Which revision is strongest?

Target word: measurable goal **Decides:** A goal must state an observable, standards-aligned outcome.

A	Correct	Observable, measurable, and tied to a grade-level standard.
B	Dropped element	Drops an observable behavior; “understand” cannot be measured directly.
C	Adjacent category	A disposition, not a measurable content goal.
D	Confusion set	Describes an activity, not a learning outcome.

3. Answer B Concept 3

A candidate reports that her class average rose from 45% to 88% in a place-value unit. ~~Her principal praised the unit at a staff meeting~~. The pre-test asked students to name the value of digits; the post-test asked them to round numbers to the nearest hundred. What is the main problem with her growth claim?

Target word: validity of growth claim **Decides:** The two tests measured different skills.

A	Confusion set	A high score is not a flaw in itself.
B	Correct	Growth needs comparable measures of the same goal.
C	Adjacent category	Aligned measures are needed for everyone, not just a subgroup.
D	Camouflage	Praise is not evidence of learning.

4. Answer D Concept 3 · transfer

A candidate wants to show growth on the goal “Students will solve two-step word problems using addition and subtraction within 100.” Which post-assessment best matches a pre-assessment of five such word problems?

Target word: design the aligned post-assessment **Decides:** Same goal, comparable format and difficulty.

A	Adjacent category	Measures attitude, not skill.
B	Dropped element	Drops the word-problem and two-step parts of the goal.
C	Wrong stem number	Raises the number range, so the task no longer matches the goal.
D	Correct	Measures the same skill in a comparable way.

5. Answer A Concept 4 • transfer

A candidate's pre-assessment shows that 7 of 22 students already mastered the unit's multiplication goal. What is the best design decision?

Target word: use pre-data to adjust design **Decides:** Some students already met the goal.

A	Correct	Pre-data changes grouping and entry points.
B	Adjacent category	Peer tutoring can help, but the 7 also need their own learning goals.
C	Absolute language	15 students still need it.
D	Dropped element	Collects data but drops its use.

6. Answer D Concept 4

Before a measurement unit, a candidate's pre-assessment shows that most students measure from the end of the ruler rather than from the zero mark. ~~She had bought new rulers with a grant from a parent group.~~ How should this finding shape her unit design?

Target word: use pre-data to adjust design **Decides:** A shared misconception is identified before instruction.

A	Absolute language	The misconception is teachable now.
B	Camouflage	Changes materials, not understanding.
C	Dropped element	Drops the pre-assessment's purpose.
D	Correct	Pre-data sets the entry point for instruction.

7. Answer B Concept 5

Halfway through a third-grade multiplication unit, a candidate's exit tickets show that a group of five students still counts by ones on arrays. The district pacing guide, ~~printed in blue~~, says to begin division next week. What decision best reflects evidence-based decision-making?

Target word: mid-unit adjustment **Decides:** Formative evidence shows a group is not progressing.

A	Dropped element	Drops the evidence.
B	Correct	Changes instruction for the group the evidence identifies.
C	Adjacent category	Over-corrects; most students are on track.
D	Confusion set	More of the same practice without a change in approach.

8. Answer C Concept 6

A candidate's class of 20 averaged 52% before and 81% after a geometry unit. ~~The unit was taught in a portable classroom.~~ Her English learners averaged 30% before and 72% after. What does the subgroup analysis show?

Target word: interpret subgroup gains **Decides:** Compare gains: class +29 points; English learners +42 points.

A	Dropped element	Drops the starting point and looks only at final scores.
B	Absolute language	Averages can hide groups that did not grow.
C	Correct	Gain, not just the final score, is the key evidence.
D	Wrong stem number	81% is the post-test average, not the gain.

9. Answer C Concept 6 • transfer

A candidate's class average rose from 60% to 85%, but one student went from 70% to 65%. How should the analysis treat this student?

Target word: individual-level analysis **Decides:** One student declined while the class grew.

A	Dropped element	Dropping the case loses important evidence.
B	Inverted operation	Overreacts; the class as a whole grew.
C	Correct	Individual analysis explains exceptions the average hides.
D	Absolute language	The average does not describe every student.

10. Answer D Concept 7

During the defense, a panelist asks why only half the class met the unit's goal on elapsed time. ~~The candidate's classroom had a clock that ran slow.~~ Which reflection is strongest?

Target word: evidence-based reflection **Decides:** Reflection should tie results to instructional causes with evidence.

A	Camouflage	Blames an outside factor without evidence.
B	Confusion set	General praise, not reflection on the results.
C	Adjacent category	Blames learners instead of analyzing instruction.
D	Correct	Names an evidence-backed cause and a specific change.

11. Answer C Concept 7 • transfer

A candidate's reflection says: "Next time I will add more manipulatives." What is missing that would make it evidence-based?

Target word: what a reflection needs **Decides:** Evidence and a causal link.

A	Adjacent category	More ideas without evidence are still unsupported.
B	Confusion set	Judgment without evidence.
C	Correct	Links the change to evidence about learning.
D	Absolute language	Reflection must connect the change to results.

12. Answer A Concept 8

A candidate ~~who drives 40 minutes to campus~~ found that her mid-unit questioning mostly asked for answers rather than explanations. A panelist asks for a professional growth goal. ~~Her mentor enjoys gardening.~~ Which goal is strongest?

Target word: growth goal from TWS evidence **Decides:** The weakness identified is low-level questioning.

A	Correct	Tied to the identified weakness, with concrete steps and evidence.
B	Adjacent category	A worthwhile goal, but not the weakness the TWS revealed.
C	Absolute language	Quantity of workshops is not a targeted goal.
D	Dropped element	Drops specifics and any measure of progress.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	1	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Contextual factors drive planning	Only the contextual factors that affect learning (readiness, language, needs, resources) belong in the plan, and each should be tied to a specific instructional decision.	Listing background facts that never change what the teacher does.	“How did your classroom context shape the unit?”
2	Measurable, standards-aligned learning goals	A learning goal states what students will be able to do, in observable terms, aligned to a standard.	Stating an activity or a vague outcome (“understand,” “appreciate”) as the goal.	Wording a goal for a K-6 math unit.
3	Aligned pre- and post-assessment	Pre- and post-assessments must measure the same goals in comparable ways so growth can be claimed.	Comparing a pre-test and a post-test that measure different skills or formats.	A claim of growth from two assessments.
4	Pre-assessment informs design	Pre-assessment results should change grouping, pacing, and entry points before instruction begins.	Giving the pre-test and then teaching the planned unit unchanged.	“What did you do with your pre-assessment data?”
5	Instructional decision-making mid-unit	When formative evidence shows a group is not progressing, the teacher changes the approach and can explain why.	Staying on the pacing guide regardless of evidence.	A mid-unit check that shows a gap.
6	Analysis at whole-class, subgroup, and individual levels	Learning analysis looks at the whole class, a meaningful subgroup, and individual students, using gain as well as final scores.	Reporting only the class average, or judging success by final scores alone.	Pre/post data and “what does it show?”
7	Evidence-based reflection	Reflection links results to specific instructional causes and names what to change, supported by evidence.	Blaming outside factors or offering general praise without evidence.	“Why do you think this worked or didn’t?”
8	Professional growth goals	A professional development goal grows from an identified weakness in the TWS and names concrete steps and evidence of progress.	Setting a goal unrelated to the portfolio’s findings.	“How will you apply what you learned?”

Course scope (public description): Mathematics (K-6) Portfolio Oral Defense: Mathematics (K-6) Portfolio Defense focuses on a formal presentation. The student will present an overview of their teacher work sample (TWS) portfolio discussing the challenges they faced and how they determined whether their goals were accomplished. They will explain the process they went through to develop the TWS portfolio and reflect on the methodologies and outcomes of the strategies discussed in the TWS portfolio. Additionally, they will discuss the strengths and weaknesses of those strategies and how they can apply what they learned from the TWS portfolio in their professional work environment.